Assignments Overdue: You can still pass! Remember, you need to pass these assignments before the course ends on July 9, 11:59 PM PDT.

THIS WEEK'S FORUM

Week 1

Discuss this week's modules here.

Go to forum

Statistical modeling and Monte Carlo estimation

Matthew Heiner

Statistical modeling, Bayesian modeling, Monte Carlo estimation

Learning Objectives

Understand how to interpret and specify the components of Bayesian statistical models: likelihood, prior, posterior.

Given a scientific problem and data, identify an appropriate likelihood/model and relevant hypotheses.

More

Use R to generate Monte Carlo samples from posterior distributions to obtain: point estimates, interval estimates, posterior probabilities of hypotheses.

Module Overview

- Course introduction 5 min
- Module 1 assignments and materials 3 min

1. Statistical Modeling

- Objectives 7 min
- Modeling process 8 min
- **Quiz:** Lesson 1 8 questions Due June 4, 11:59 PM PDT
- **Discussion Prompt:** Statistical modeling process 15 min

2. Bayesian Modeling

- Components of Bayesian models 8 min
- Model specification 7 min
- Posterior derivation 9 min
- Non-conjugate models 7 min
- **Quiz:** Lesson 2 8 questions Due June 4, 11:59 PM PDT
- Reference: Common probability distributions

3. Monte Carlo Estimation

- Monte Carlo integration 9 min

Monte Carlo error and marginalization 6 min

Computing examples 15 min

Computing Monte Carlo error 13 min

Quiz:
Lesson 3 8 questions Due June 4, 11:59 PM PDT

Code for Lesson 3

Background for Lesson 4

Markov chains 20 min

Optional Honors Content

Markov Chains

Quiz:
Markov chains 5 questions Due June 4, 11:59 PM PDT